

A Reusable Semantic Web Framework for Evidence-Grounded Fundamental Rights Impact Assessments under the EU AI Act

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The EU AI Act (Art. 27) requires deployers of high-risk AI systems to conduct Fundamental Rights Impact Assessments (FRIAs) before deployment, yet the evidence needed for credible assessments is fragmented across incompatible incident repositories, risk vocabularies, and legal texts. We present a reusable Semantic Web-based framework that consolidates this evidence for two high-risk public sector categories: employment and worker management (Annex III(4)) and access to essential public services (Annex III(5)(a)). A curated 150-record corpus is annotated along four axes using keyword, LLM, and hybrid methods and serialised as a SPARQL-queryable knowledge graph of 1,351 RDF triples. Five FRIA demonstration scenarios surface 103 records (68.7% coverage). Evaluation against a 69-record gold standard reveals that LLM-assisted classification of the employment domain achieves only $\kappa = 0.045$, a cautionary result for automated fairness-related evidence retrieval in this domain. All artefacts are released openly to support adoption by regulators, national authorities, and SMEs.

Keywords: EU AI Act, Fundamental Rights Impact Assessment, knowledge graph, Semantic Web, LLM classification, algorithmic fairness, public sector AI

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1 Introduction

Government agencies across Europe are deploying large language models (LLMs) at scale: welfare eligibility triage, citizen-facing chatbots, recruitment screening pipelines, and caseworker decision support [4]. The appeal is straightforward, but the risks to fundamental rights are direct. Bias amplification, hallucinated outputs, and opacity bear on individuals' access to work and essential services in ways that are not always recoverable [3, 26].

Real-world cases demonstrate what inadequate assessment looks like in practice. In the Netherlands, the Tax Administration's algorithmic risk scoring system embedded racial profiling and operated without meaningful oversight, wrongly accusing tens of thousands of families of benefit fraud [1]. The SyRI case established that automated profiling in public welfare contexts can constitute a violation of the European Convention on Human Rights [7, 27]. In employment, the *Mobley v. Workday* litigation alleges that an AI-powered résumé screening tool systematically discriminates on the basis of age, race, and disability. A persistent pattern runs through these incidents: high-stakes public sector AI can cause large-scale rights violations when risk assessment is inadequate.

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The EU AI Act responds to this directly. Article 27 requires that deployers of public sector high-risk systems conduct a Fundamental Rights Impact Assessment before deployment, documenting how the system may affect rights including non-discrimination, privacy, and good administration [8]. The European Union Agency for Fundamental Rights has noted that many deployments still lack the tools and evidence base to conduct meaningful assessments [10].

The evidence deployers need does not currently exist in a usable form. Incident data resides in repositories such as AIAAIC and AIID, which use free-text narratives and community-developed tags rather than AI Act terminology [17, 22]. Semantic Web risk vocabularies—DPV, VAIR, and AIRO—provide structured, machine-readable taxonomies aligned with the regulatory framework but are not integrated into FRIA workflows [15, 25]. Legal obligations in Annex III and the EU Charter of Fundamental Rights are expressed in natural language that does not support automated processing [9]. Deployers must manually reconcile these disparate sources with no reusable infrastructure to connect incident evidence, risk vocabularies, and regulatory obligations.

This paper presents a framework that bridges that gap. The full technical implementation is described in the accompanying dissertation [20]. This extended abstract presents the framework, its key empirical findings, and their implications for algorithmic fairness research and practice.

2 Framework

2.1 Corpus Construction

The corpus combines three complementary evidence sources: approximately 100 records from the AIAAIC incident repository [22], 30 entries from the U.S. Federal AI Use Case Inventory [19], and 20 European Court of Human Rights cases from HUDOC [6]. Together these provide 150 records covering LLM-related and analogous AI deployments in employment and essential public services. The three sources triangulate across evidence types: AIAAIC captures real-world failures; the U.S. Federal Inventory documents active government deployments; ECtHR cases provide authoritative legal analysis of rights violations.

Each record is classified along four axes: (1) Annex III domain (employment or essential services); (2) implicated EU Charter rights (multi-label); (3) risk pattern (bias/discrimination, privacy breach, procedural unfairness, lack of transparency, or other); and (4) causal factor (data quality, model design, deployment context, or oversight failure).

2.2 Semantic Schema and Knowledge Graph

The schema is grounded in Semantic Web standards and aligned with DPV [21, 25], VAIR [15], AIRO [12], and the FRIA ontology [24]. Reusing established vocabularies reduces the learning burden for adopters and maximises interoperability with existing compliance tooling. The annotated corpus is serialised in two formats: a Turtle RDF knowledge graph of 1,351 triples (193 nodes, 965 edges) enabling SPARQL queries, and 150 JSON-LD records for web-compatible consumption.

2.3 Annotation Pipeline

Three annotation methods are implemented and compared. A keyword baseline uses synonym-expanded dictionary matching against controlled vocabularies. An LLM classifier applies Claude Sonnet [2] with zero-temperature sampling, few-shot prompting, and structured output constraints. A hybrid method resolves conflicts between

keyword and LLM outputs using priority logic, retaining keyword classifications for domain and LLM classifications for rights where each method performs better.

2.4 Regulatory Crosswalk

A regulatory crosswalk maps the obligations arising from Annex III(4) and Annex III(5)(a) to specific articles of the EU Charter of Fundamental Rights [9]. This mapping is not enumerated in the AI Act itself; the crosswalk makes it explicit and machine-readable, reducing interpretive uncertainty for deployers and providing the structured link between regulatory requirements and rights evidence that the schema’s multi-label rights axis builds on.

2.5 Query Interface

A Flask web application loads the knowledge graph on startup and allows users to filter records by Annex III domain, Charter rights, and risk pattern through a form-based interface, moving the framework toward the interactive compliance tooling anticipated by Art. 27(5) [8].

3 Evaluation

3.1 Gold Standard and Agreement Metrics

Of the 150 corpus records, 69 (46%) were manually annotated to form a gold standard, stratified across sources and both Annex III domains. Each record was annotated by reviewing original source material rather than summary fields alone, following written guidelines with explicit decision criteria per axis. Cohen’s κ [5] and percentage agreement are reported for each axis and method, interpreted using the Landis and Koch scale [16].

3.2 Domain Classification

For the essential services domain, the best-performing method achieves $\kappa = 0.525$ (moderate agreement). For employment, the best-performing method achieves $\kappa = 0.045$ —near-chance. This divergence is attributable to spurious vocabulary correlations: a disproportionate share of records discuss employment incidentally rather than as the primary domain of harm, and both keyword and LLM classifiers over-predict employment for records containing general labour market language.

The implication for algorithmic fairness research is direct. Employment discrimination is among the most extensively documented harms of automated decision-making, and Annex III(4) covers the AI Act’s highest-risk employment applications. Near-chance agreement means that automated annotation of employment-domain risk evidence cannot currently be trusted without human review. Deployers relying on such tools to surface employment risk evidence for a FRIA risk generating an evidence base that is both incomplete and misleading. The finding holds across models: a comparison with GPT-4o-mini yields $\kappa = 0.196$ for domain classification, confirming the difficulty lies in the task rather than in a single model.

3.3 Coverage and Risk Patterns

Five FRIA demonstration scenarios query the populated knowledge graph for records relevant to realistic deployer contexts: welfare eligibility AI, public sector recruitment screening, surveillance in public housing, LLM decision-support for caseworkers, and a cross-domain thematic review by a national regulator. Across all five, 103 of the

150 records are surfaced (68.7% coverage). The 31.3% not surfaced are predominantly records with unknown classifications produced by pipeline underclassification—the direct consequence of the domain agreement failures described above.

The hybrid method reduces unknown risk pattern classifications from 61.3% (keyword baseline) to 14.0%, substantially enlarging the evidence base available for compliance queries. Agreement on risk patterns remains in the slight-to-fair range, with procedural unfairness and lack of transparency the categories of greatest disagreement, driven primarily by multi-label harm structures where incidents involve co-occurring failure modes.

4 Limitations

Four limitations bear on the framework’s current scope. The gold standard was produced by a single annotator; multi-annotator reliability using Fleiss’ κ [11] is a necessary next step before pipeline outputs can be used without human review. The corpus covers only two of the eight Annex III categories; the schema is designed for extensibility but coverage of biometrics, law enforcement, and migration is not yet tested, a gap that prior work on high-risk AI classification highlights [13, 14]. AIAAIC is predominantly English-language and draws heavily on U.S. and U.K. sources, introducing geographic bias in an EU regulatory context. Finally, employment classification quality is insufficient for unsupervised deployment and requires human oversight for any FRIA relying on that domain’s evidence.

5 Relevance to ECAF

This work contributes across three of ECAF’s disciplinary areas.

Policy and Law. The regulatory crosswalk and FRIA demonstration scenarios directly address impact assessment obligations under the AI Act, providing concrete, reusable infrastructure for Art. 27 compliance and contributing to emerging standardisation of AI incident reporting [18]. The analysis of GDPR versus AI Act impact assessment requirements [23] provides complementary legal framing.

Computer Science. The annotation pipeline, multi-method comparison, and gold standard evaluation contribute empirical evidence on the reliability of LLM-assisted classification in a regulatory context, with direct implications for auditing framework design.

Social Sciences. The corpus documents real-world AI failures in employment and essential services, two domains where automated systems bear most directly on the rights of marginalised groups. The near-chance employment domain agreement ($\kappa = 0.045$) is a cautionary result for practitioners building automated fairness assessment tools in this domain.

6 Conclusion

We have presented a Semantic Web-based framework that consolidates fragmented AI risk evidence into a structured, SPARQL-queryable knowledge graph supporting EU AI Act FRIA compliance. Five FRIA demonstration scenarios achieve 68.7% coverage of a 150-record corpus, and gold standard evaluation reveals that LLM-assisted employment domain classification is near-chance—a result with direct implications for automated fairness assessment in the domain most associated with algorithmic discrimination. All schema artefacts and annotated data are released under CC BY 4.0 and pipeline source code under the MIT Licence at <https://github.com/faitholopade/Dissertation>,

to support adoption by regulators, national authorities, and SMEs extending the framework to further Annex III categories.

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